INSTITUTE	**FACULTY OF PHARMACY**
PROGRAM	**BACHELOR OF PHARMACY**
SEMESTER	**3**
COURSE TITLE	**PHARMACEUTICAL ORGANIC CHEMISTRY-II**
COURSE CODE	**13PH0301**
COURSE CREDITS	**6**

Objective:

1 Write the structure, name and the type of isomerism of the organic compound
2 Write the reaction, name the reaction and orientation of reactions
3 Account for reactivity/stability of compounds,
4 Prepare organic compounds
5 The syllabus emphasizes on mechanisms and orientation of reactions. Chemistry of fats and oils are also included in the syllabus.

Course Outcomes: After completion of this course, student will be able to:

1 Understand mechanisms and orientation of chemical reactions.
2 Acquire basic knowledge regarding general methods of preparation of organic compounds.
3 Understand the reactions and reactivity of some organic compounds and Apply knowledge of heterocyclic compounds, electrophilic and nucleophilic reactions.
4 Understand the chemistry of fats and oils.

Pre-requisite of course: The syllabus emphasizes on mechanisms and orientation of reactions. Chemistry of fats and oils are also included in the syllabus.

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	1	4	75	15	10	35	15

Contents : Unit	Topics	Contact Hours
1	**Benzene and its derivatives:** Analytical, synthetic and other evidences in the derivation of structure of benzene, Orbital picture, resonance in benzene, aromatic characters, Huckel's rule. ? Reactions of benzene - nitration, sulphonation, halogenation - reactivity, Friedelcrafts alkylation - reactivity, limitations, Friedelcrafts acylation. ? Substituents, effect of substituents on reactivity and orientation of mono substituted benzene compounds towards electrophilic substitution reaction. ? Structure and uses of DDT, Saccharin, BHC and Chloramine.	10

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Contents : Unit	Topics	Contact Hours
2	**Phenols** Acidity of phenols, effect of substituents on acidity, qualitative tests, Structure and uses of phenol, cresols, resorcinol, naphthols. Aromatic Amines: Basicity of amines, effect of substituents on basicity, and synthetic uses of aryl diazonium salts. Aromatic Acids: Acidity, effect of substituents on acidity and important reactions of benzoic acid.	10
3	**Fats and Oils:** Fatty acids – reactions. Hydrolysis, Hydrogenation, Saponification and Rancidity of oils, Drying oils. Analytical constants – Acid value, Saponification value, Ester value, Iodine value, Acetyl value, Reichert Meissl (RM) value – significance and principle involved in their determination.	10
4	**Polynuclear hydrocarbons:** ? Synthesis, reactions. ? Structure and medicinal uses of Naphthalene, Phenanthrene, Anthracene, Diphenylmethane, Triphenylmethane and their derivatives.? Synthesis, reactions. ? Structure and medicinal uses of Naphthalene, Phenanthrene, Anthracene, Diphenylmethane, Triphenylmethane and their derivatives.	8
5	**Cyclo alkanes:** Stabilities – Baeyer's strain theory, limitation of Baeyer's strain theory, Coulson and Moffitt's modification, Sachse Mohr's theory (Theory of strain less rings), reactions of cyclopropane and cyclobutane only.	7
Total Hours		**45**

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
1	**Practical** Experiment no.1, Experiment no.2, Experiment no.3, Experiment no.4, Experiment no.5, Experiment no.6, Experiment no.7, Experiment no.8, Experiment no.9, Experiment no.10, Experiment no.11, Experiment no.12, Experiment no.13	60
2	**Tutorial** Workshop 1, Workshop 2, Workshop 3, Workshop 4, Workshop 5, Workshop 6, Workshop 7, Workshop 8, Workshop 9, Workshop 10, Workshop 11, Workshop 12, Workshop 13, Workshop 14, Workshop 15	15
Total Hours		**75**

Textbook :

1 Organic Chemistry , Morrison and Boyd, Pearson Education India, 2010

References:

1 Organic Chemistry Vol. 1 , Organic Chemistry Vol. 1 , I.L. Finar, Pearson Education India, 2002

2 A Textbook of Organic Chemistry , A Textbook of Organic Chemistry , B. S. Bahl & Arun Bahl. , S Chand & Company, 2016

3 Textbook of Organic Chemistry , Textbook of Organic Chemistry , P. L. Soni, Sultan Chand & Sons, 2012

4 Practical Organic Chemistry, fourth edition, Practical Organic Chemistry, fourth edition, Mann and Saunders. , Pearson Education India, 2009

5 Vogel's Textbook of Practical Organic Chemistry, 5 edition, Vogel's Textbook of Practical Organic Chemistry, 5 edition, Brian S. Furniss, Pearson India, 2003

6 Advanced Practical Organic Chemistry, Advanced Practical Organic Chemistry, N. K. Vishnoi, S Chand, 2010

7 Introduction to Organic Laboratory Techniques: A Small-Scale Approach, International Edition, Introduction to Organic Laboratory Techniques: A Small-Scale Approach, International Edition, Donald L. Pavia, Brooks/Cole, 2010

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery					
Remember / Knowledge	**Understand**	**Apply**	**Analyze**	**Evaluate**	**Higher order Thinking / Creative**
20.00	30.00	25.00	15.00	10.00	0.00

Instructional Method:

1 The course delivery method will depend upon the requirement of content and the need of students. The teacher in addition to the conventional teaching method by the blackboard may also use any tools such as demonstration, role play, quiz, brainstorming, MOOCs etc.

2 The internal evaluation will be done based on continuous evaluation of students in the laboratory and classroom.

3 Students will use supplementary resources such as online videos, NPTEL videos, MOOCs/ e-courses, virtual laboratories.

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